

Supplementary Information for Metadata Reliance in Vision-Language Models for CT Lung Nodule Malignancy Prediction

This compact supplement contains Supplementary Notes 1–8 and Supplementary Tables 1–3. Reporting checklists are submitted as separate checklist files. Full per-case predictions, long subgroup tables, calibration tables, decision-curve outputs, run manifests, and source code are provided in the anonymized review repository and will be linked publicly on acceptance.

Supplementary Note 1. Anonymization and Leakage Prevention

The LUNA25 benchmark was used as a retrospective internal-control setting to test whether hosted VLMs could use nodule-centered CT crops and structured clinical context as malignancy classifiers. Inference prompts did not expose patient identifiers, scanner identifiers, study dates, file names, dataset split metadata, target labels, or annotation identifiers. Clinical text was generated from released tabular descriptors and did not include outcomes. The 20-shot exemplars were fixed before test-set evaluation and reused unchanged across conditions. LNDb cases were never used as in-context exemplars, calibration samples, or prompt-adaptation cases.

Supplementary Note 2. Prompt/Input Condition Definitions

The Z/F labels describe model-independent prompt/input conditions rather than a single vendor. The complete six-condition video ablation was performed with Gemini 3 Flash Preview. Cross-family rows used the same Z2, Z3, and Z0 definitions under a standardized sparse montage.

Condition	Definition
Z0	Zero-shot metadata-only control: rich radiology prompt plus structured clinical text, no CT image/video and no exemplar labels/text. Used in the cross-family sparse-montage audit.
Z1	Zero-shot CT-video only with a minimal system prompt.
Z2	Zero-shot CT-video or sparse-montage image only with the rich radiology prompt and no structured clinical text.
Z3	Zero-shot CT-video or sparse-montage image plus structured clinical text with the rich radiology prompt.
F0	Matched F3 text-only control: same rich scaffold, exemplar labels/text, target clinical text, JSON schema, model identifier, and temperature as F3, but all CT videos were removed.
F1	20-shot CT-video only with the minimal system prompt.
F2	20-shot CT-video only with the rich radiology prompt.
F3	20-shot CT video plus structured clinical text with the rich radiology prompt; primary metadata-assisted VLM condition.

For few-shot conditions, each exemplar contributed one user turn containing the exemplar video and, when applicable, exemplar clinical text, followed by an assistant turn containing the desired JSON response with a high or low exemplar probability. Ten malignant exemplars followed ten benign exemplars in the fixed order stored in the repository. The target case was supplied only in the final user turn and the response was constrained to JSON containing a malignancy **confidence** score and one-sentence **reasoning**.

Supplementary Note 3. CT-to-Video and Sparse-Montage Representation

Each target case was represented as a nodule-centered axial CT crop of $64 \times 128 \times 128$ voxels. Hounsfield-unit values were clipped to $[-1000, 1000]$, linearly rescaled to 8-bit, triplicated to RGB, and encoded as 20-fps MP4 video. The prompt explicitly stated that the nodule was centered in each frame and asked for a JSON malignancy score.

A hosted Gemini video-sampling probe indicated that the interface effectively sampled approximately three frames from the 64-frame clip, with nominal sampled positions corresponding to frames 9, 29, and 49. The cross-family audit therefore used a standardized axial montage of these three frames. This montage made the visual input portable across hosted model families but was not assumed to be pixel-token equivalent to native video ingestion, because the three frames were concatenated into a single image and could undergo different resizing, layout parsing, and

modality-specific preprocessing.

Supplementary Note 4. Structured Clinical Text and Field Provenance

Clinical text was generated from released tabular fields using a fixed template. If a descriptor was unavailable, the corresponding sentence was omitted; coded “not determined” categories were retained verbatim. A representative target text was:

A 55-year-old White male patient who is a current smoker presents with a pulmonary nodule. Located in the right upper lobe. With a long-axis diameter of 52.0 mm and perpendicular diameter of 39.0 mm. Classified as a solid nodule. Exhibiting lobulated margins. With soft tissue attenuation.

The imaging crops and binary malignancy labels came from the LUNA25 release. Other demographic, smoking-history, and CT-descriptor variables were joined from the matched public NLST records. The `sct_*` variables are reader-derived structured CT annotations rather than scanner-native metadata or new annotations created for this study.

Supplementary Table 1. Provenance and availability of structured fields used in Z3/F3 text and secondary analyses.

Field	Provenance / interpretation	Available in LUNA25 test split
Age_at_StudyDate	Public demographic field; used as age.	917/917
Gender	Public demographic field; used as sex.	917/917
race	Self-reported NLST category; used verbatim in Z3/F3 text.	917/917
cigsmok	Public screening-history field; used as smoking status.	917/917
sct_long_dia	Reader-derived long-axis CT diameter.	701/917
sct_perp_dia	Reader-derived perpendicular CT diameter.	701/917
sct_epi_loc	Reader-derived lobar location descriptor.	706/917
sct_margins	Reader-derived margin category.	706/917
sct_pre_att	Reader-derived attenuation/density category.	706/917
label	Curated LUNA25 binary malignancy label.	917/917

Supplementary Note 5. Hosted Model Interfaces and Inference Settings

Hosted VLM inference for the reported VLM rows was performed on April 30, 2026 using temperature 0 where supported. Model identifiers, prompt templates, serialized requests, and parsed JSONL outputs are included in the anonymized review repository. The cross-family audit was designed to test whether metadata reliance generalized across hosted VLM interfaces under a portable sparse visual input, not to rank vendor-native image or video ingestion. GPT-5.5 image-only outputs were constant 0.5 with no-image rationales in the tested interface and were retained only as an interface-provenance check.

Supplementary Table 2. Hosted VLM rows and input route.

Model row	Input route and manuscript role
Gemini 3 Flash Preview	Native hosted video for Z1–Z3/F1–F3/F0; sparse montage bridge for Z2/Z3/Z0. Primary six-condition ablation and bridge to the cross-family audit.
Gemini 3.1 Pro Preview	Sparse montage under Z2/Z3/Z0; F3 generation-context comparison. Cross-family metadata-reliance audit and generation sensitivity.
Claude Opus 4.8	Sparse montage under Z2/Z3/Z0. Cross-family metadata-reliance audit.
GPT-5.5	Sparse montage under Z2/Z3/Z0, with image-only delivery failure in the tested interface. Metadata-assisted and text-only rows reported; image-only row used only for interface provenance.
Gemini 2.5 Flash/Pro and MedGemma 1.5-4B	F3 only. Contextual model extensions, not primary contrasts.

Supplementary Note 6. Rationale-Text Overlap Audit

We performed a deterministic text-overlap audit to quantify how often the one-sentence Gemini F3 rationales echoed structured descriptors supplied in the clinical text. This audit measures textual reuse, not causal feature reliance, because generated rationales are not guaranteed to be faithful explanations of predictions. A rationale was scored as repeating diameter when it contained the supplied long-axis diameter in integer or one-decimal millimetre form and as repeating margin when it contained the supplied margin term (`spiculated`, `irregular`, `lobulated`, `smooth`, or `well-defined`).

	Rationale text repeated the supplied...	Fraction of F3 outputs
	Nodule diameter	504/917 (55.0%)
	Nodule diameter among rows with recorded diameter	504/701 (71.9%)
	Margin descriptor	637/917 (69.5%)
	Diameter or margin descriptor	690/917 (75.2%)

These results support the quantitative controls showing that Z3/F3 and F0 align more strongly with structured descriptors and Brock probability than visual-only conditions. The audit is reproduced by `benchmark/rationale_audit.py`.

Supplementary Note 7. Failure-Mode Lexicon

Failure-mode coding was performed on the representative single Gemini F3 run at the 0.5 decision threshold. Each discordant case’s rationale was classified with a fixed, case-insensitive substring lexicon:

- **Margin/spiculation:** `spicul`, `margin`, `lobulat`, `irregular`, `ill-defined`, `well-defined`, `smooth`, `border`, `contour`.
- **Attenuation:** `solid`, `part-solid`, `subsolid`, `ground-glass`, `ggo`, `attenuat`, `density`, `soft tissue`, `calcif`.
- **Patient history/smoking:** `smok`, `history`.
- **Size:** `size`, `diameter`, `sized`, `large`.

At threshold 0.5, Gemini F3 misclassified 280/917 cases: 258 false positives and 22 false negatives. Among false positives, rationales most often cited margin/spiculation (192/258, 74.4%), attenuation (180/258, 69.8%), history/smoking (149/258, 57.8%), and size (63/258, 24.4%). Every false-negative rationale cited margin/spiculation (22/22), with history/smoking in 12/22 and size in 11/22. The representative discordant cases are shown in main-text Figure 4. The lexicon is reproduced by `benchmark/failure_mode_lexicon.py`.

Supplementary Note 8. Exploratory Analyses and Repository Outputs

Calibration, decision-curve, PR-AUC, subgroup, race-excluded structured-model, F3 repeat-averaging, dense-sampling, open-weight Gemma, and Med3D ResNet18+TTT analyses were exploratory and not adjusted for multiplicity. They are retained as reproducibility artifacts rather than long printed supplementary tables. The exploratory ResNet18+TTT wrapper placed a TTT block between the Med3D ResNet-18 layer-4 representation and a LayerNorm–Dropout–Linear classifier head; TTT type, training data size, and optimization scope were crossed in a 12-row grid and are reported in the released metrics CSV. These experiments were not run on LNDb and should not be interpreted as an externally validated eighth baseline.

Supplementary Table 3. Key reproducibility artifacts.

Artifact	Contents
<code>results/vlm/</code>	VLM JSONL outputs, F3 repeat files, F0 text-only control, cross-family audit outputs, and Brock predictions.
<code>data/predictions/luna25_dl/files/</code>	LUNA25 supervised DL prediction CSVs.
<code>data/predictions/lndb_dl/</code>	LNDb supervised DL prediction CSVs.
<code>benchmark/canonical_numbers.py</code>	Regenerates the AUCs reported in the manuscript tables and figures from per-case files.
<code>benchmark/calibration_and_subgroup.py</code>	Generates calibration, PR-AUC, decision-curve, subgroup, and post-hoc operating-point summaries.
<code>results/figures/</code>	Machine-readable tables underlying the manuscript and repository figures, including calibration and subgroup CSVs.
<code>results/ttt/med3d_resnet18_adaptation_metrics.csv</code>	Full 12-row exploratory Med3D ResNet18+TTT metrics grid.
<code>benchmark/few_shot_samples.json</code> and <code>data/metadata/clinical_texts.csv</code>	Fixed exemplar identifiers and structured clinical-text source table.
<code>requirements.txt</code>	Pinned Python package versions and runtime requirements for the released analysis code.